

Confidence Intervals and the Z-procedure

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Where this fits: Pipeline map step 0 (Foundations). See the [Course map](#).



- **Builds on:** Standard normal and the z-table ([Note 251](#)); Central limit theorem ([Note 272](#)).
- **Compare with:** Hypothesis testing: null and alternative ([Note 290](#)).

1. Overview

Key point: A confidence interval replaces a single guess with a range: point estimate $\pm$ margin of error, together with a confidence level such as 95%.

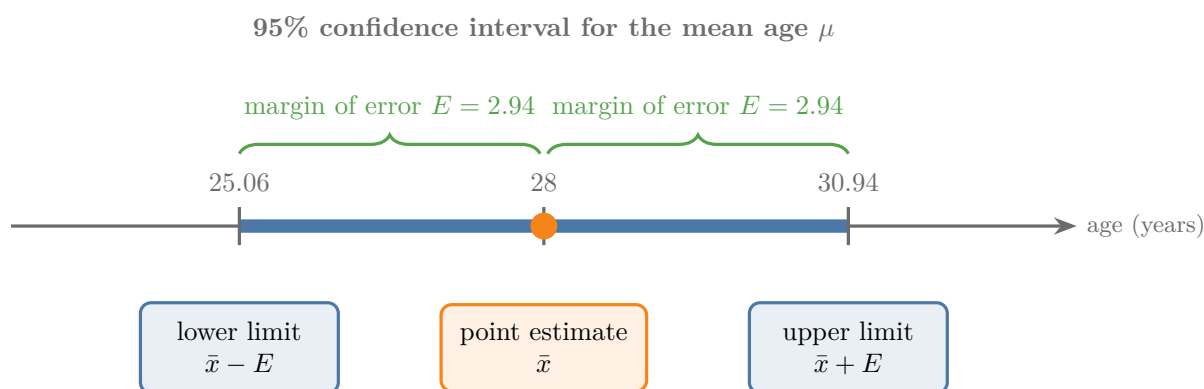


Figure 1: A 95% confidence interval for a mean age: the point estimate in the middle, the margin of error on each side

A sample mean is almost never exactly the population mean. A **confidence interval** admits this: it gives a range of plausible values for the parameter and says how confident we are in the method that built it. Figure 1 shows its parts for one example worked out in this Note.

This Note covers:

- why a point estimate alone is not reliable;
- what a confidence interval and a confidence level are;
- the **z-procedure**, used when the population standard deviation σ is known: its assumptions, its formula, where the formula comes from, and how to find the critical value $z_{\alpha/2}$.

The [interpreting confidence intervals Note](#) explains what “95% confident” really means and what makes an interval wide or narrow. The [t-procedure Note](#) handles the usual case where σ is unknown.

2. Terms we build on

Key point: Population and sample, parameter and statistic, inferential statistics, the central limit theorem and the point estimate.

Each of these is taught in an earlier Note; here is the one-line version.

- **Population and sample:** the population is the entire group we want to study; a sample is the part we measure, chosen at random and representative (see the [what is statistics Note](#)).
- **Parameter and statistic:** a number describing the population (Greek letters: μ, σ) and the same number computed from a sample (Latin letters: $\bar{x}, s$). Parameters are usually unknown and must be estimated from statistics.
- **Inferential statistics:** drawing conclusions about a population from a sample. Its tools include confidence intervals, hypothesis tests and regression, and it answers questions such as “can the sample mean tell us the population mean?”.
- **Central limit theorem:** the means of large samples are approximately normal around μ with standard error $\sigma/\sqrt{n}$ (see the [central limit theorem Note](#)).
- **Point estimate:** a single number computed from sample data as the best guess for an unknown parameter (see the [estimating a mean Note](#)).

The running example of this Note: an online channel has about 77,000 subscribers, and we want their average age μ . Asking all of them is impossible. Instead, 100 subscribers in an online event type their age in the chat. Their mean age is $\bar{x} = 28$ years: a point estimate of μ .

Averaging the means of 10 such classes would give a better point estimate, as in the estimating a mean Note. It would still be a single number.

3. Why a point estimate is not enough

Key point: An exact guess is almost always wrong; a range is far more likely to be right.

Suppose we must guess how many runs MS Dhoni will score in today’s match, with three ways to bet:

Bet	Wins if	Prize
Exact score	the guess is exactly right	1 crore rupees
Within ± 10 runs	the score is within 10 of the guess	75 lakh rupees
Within ± 20 runs	the score is within 20 of the guess	50 lakh rupees

A sensible person takes the second or third bet. Hitting the exact score is very unlikely, while a range of 20 or 40 runs has a real chance. If we guess 35 and he scores 28, the difference is 7: the ± 10 bet wins.

The same holds for estimates. Claiming that the mean age of 77,000 people is exactly 28, because 100 of them average 28, is not believable. So statisticians report a range that probably contains the parameter.

4. Confidence intervals and confidence levels

Key point: A confidence interval is a range of plausible values for a population parameter; the confidence level, usually 95%, says how often the method that builds such intervals succeeds.

A **confidence interval (CI)** is a range of values within which we expect a population parameter, such as μ or σ , to lie. It expresses the uncertainty of an estimate obtained from a sample.

Every confidence interval comes with a **confidence level**, a percentage such as 95%. “The mean age of the subscribers is between 25 and 32 years, at 95% confidence”: the range 25 to 32 is the interval, and 95% is the level. The exact meaning of the level is subtle; the [interpreting confidence intervals Note](#) is devoted to it.

Every confidence interval has the same structure.

1. **In words:** start from the point estimate and go one **margin of error** down and one up.
2. **Formula:**

$$\text{CI} = \text{point estimate} \pm \text{margin of error}$$

3. **Example:** a point estimate of 25 years and a margin of error of 4 years give

$$25 \pm 4 : \quad 25 - 4 = 21 \text{ (lower limit)}, \quad 25 + 4 = 29 \text{ (upper limit)}$$

So building a confidence interval needs two things: a point estimate and a margin of error. The rest of this Note is about the margin of error.

A confidence interval is always about a **parameter** of the population (here μ), never about a statistic. The statistic $\bar{x}$ is known exactly; it is the tool we use to build the interval.

4.1 Confidence intervals in practice

Key point: Business forecasts and the error bars that seaborn draws on a bar chart are confidence intervals.

Companies, finance and economics rarely report a single projection. A sales forecast is a range: “between 4.2 and 4.8 crore rupees next quarter”.

We have also drawn confidence intervals already without naming them. Figure 2 plots the mean Titanic fare of each passenger class with seaborn, as in the [bivariate analysis Note](#). The black line on top of each bar is a 95% confidence interval for the mean fare of that class.

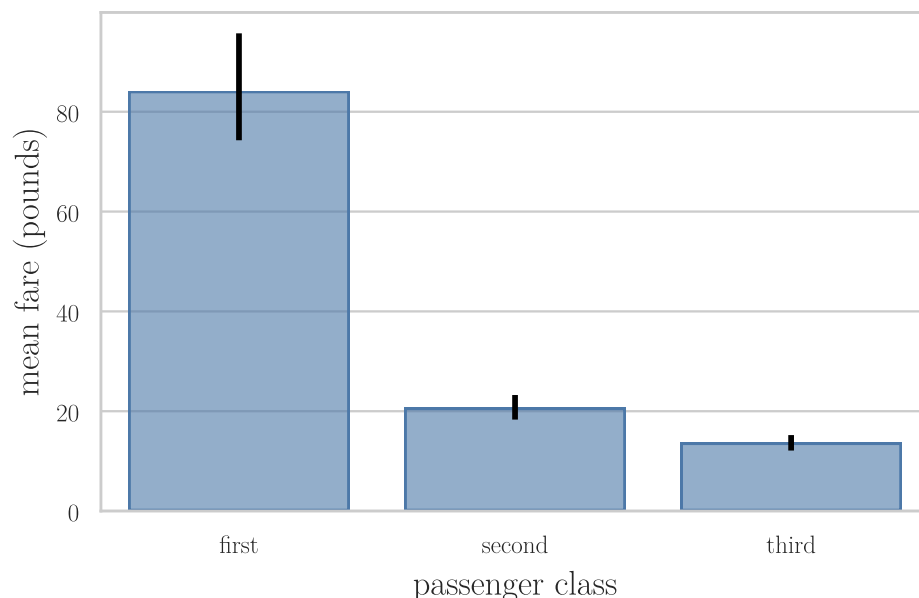


Figure 2: Mean Titanic fare per class (bars) with 95% confidence intervals (black lines)

The bars show the sample means; the lines show where the population means of passengers like these probably lie. First class has few passengers and very spread-out fares, so its interval is wide (about 74 to 95 pounds); third class has many passengers and similar fares, so its interval is narrow (about 12.6 to 14.7 pounds).

Extra: Seaborn builds these intervals by bootstrapping (see the [bagging Note](#)): it resamples the data with replacement 1000 times, recomputes the mean each time, and keeps the middle 95% of those means. For a mean from a large sample, this gives almost the same range as the formula of this Note.

5. Two ways to compute a confidence interval for a mean

Key point: The z-procedure needs the population standard deviation σ ; the t-procedure works with the sample standard deviation s .

There are several ways to compute a confidence interval. For a mean, the two standard ones are:

Procedure	Use when	Spread used
Z-procedure (“sigma known”)	the population standard deviation σ is known	σ
T-procedure (“sigma unknown”)	σ is unknown	the sample standard deviation s

In practice σ is almost never known: if we do not know the mean age of 77,000 subscribers,

we hardly know the standard deviation of their ages. So the t-procedure is the one used in real work. The z-procedure comes first because it is simpler and the t-procedure is built on it.

6. Assumptions of the z-procedure

Key point: A random sample, a known σ , and a normal population or a sample large enough for the central limit theorem.

The z-procedure gives correct intervals only when three assumptions hold:

1. **Random sample.** The sample is drawn at random and is representative. A sample of only Indian subscribers says little about a channel with subscribers in many countries.
2. **Known population standard deviation.** We know σ , here the standard deviation of the ages of all subscribers. We assume $\sigma = 15$ years.
3. **Normal sampling distribution.** Either the population itself is normal, or the sample is large, usually $n > 30$. Then, by the central limit theorem, the sample mean is approximately normal whatever the shape of the population (see the [central limit theorem Note](#)).

The third assumption is therefore mild. A population of ages with three peaks, at 26, 32 and 44, is clearly not normal; with a sample of 100 the z-procedure still works. With a non-normal population and only 20 people in the sample, it does not: neither the population nor the CLT gives a normal sample mean.

7. The z-procedure formula

Key point: $\bar{x} \pm z_{\alpha/2} \sigma / \sqrt{n}$, where $1 - \alpha$ is the confidence level and $z_{\alpha/2}$ is read from the standard normal curve.

The problem: a sample of $n = 100$ subscribers has mean age $\bar{x} = 28$ years, the population standard deviation is $\sigma = 15$ years, and we want a 95% confidence interval for μ .

1. **In words:** go from the sample mean $z_{\alpha/2}$ standard errors down and up. The standard error is $\sigma / \sqrt{n}$, and the margin of error is $z_{\alpha/2}$ times it.
2. **Formula:**

$$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

- $\bar{x}$: the sample mean, our point estimate.
- $1 - \alpha$: the confidence level. For 95%, $1 - \alpha = 0.95$, so $\alpha = 0.05$ and $\alpha/2 = 0.025$.
- $z_{\alpha/2}$: the **critical value**, the z-score that leaves an area of $\alpha/2$ in the upper tail of the standard normal curve. For 95% it is 1.96 (section 9).
- σ : the population standard deviation; n : the sample size.

3. **Example:**

$$SE = \frac{15}{\sqrt{100}} = \frac{15}{10} = 1.5, \quad E = 1.96 \times 1.5 = 2.94$$

$$28 \pm 2.94 : \quad 28 - 2.94 = 25.06, \quad 28 + 2.94 = 30.94$$

The 95% confidence interval for the mean age of all subscribers is **25.06 to 30.94 years** (Figure 1).

The formula leaves two questions: why this formula, and how to find $z_{\alpha/2}$. The next two sections answer them.

Python: The z-interval with scipy.

```
import numpy as np
from scipy import stats

x_bar, sigma, n = 28, 15, 100
se = sigma / np.sqrt(n)           # 1.5
z = stats.norm.ppf(0.975)         # 1.96
x_bar - z * se, x_bar + z * se    # (25.06, 30.94)
stats.norm.interval(0.95, loc=x_bar, scale=se)  # same
```

8. Where the formula comes from

Key point: Standardize the sample mean, find the middle 95% of the standard normal curve, and rearrange the inequality so that μ stands alone.

Step 1: the sample mean is normal. By the assumptions (or the CLT), $\bar{X}$ follows a normal distribution with mean μ and standard deviation $\sigma/\sqrt{n}$. Imagine many online events of 100 subscribers each: their mean ages $\bar{x}_1, \bar{x}_2, \dots$ form this bell (see the [central limit theorem Note](#)).

Step 2: standardize. Subtracting the mean and dividing by the standard deviation turns any normal variable into the standard normal $Z \sim N(0, 1)$ (see the [standard normal Note](#)):

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

Step 3: the middle $1 - \alpha$ of Z . We want a range of z values that contains Z with probability $1 - \alpha$ (95%). The curve is symmetric, so the remaining α splits into $\alpha/2$ in each tail (Figure 3). The two cut-off points are $-z_{\alpha/2}$ and $+z_{\alpha/2}$:

$$P(-z_{\alpha/2} < Z < z_{\alpha/2}) = 1 - \alpha$$

Step 4: put $\bar{X}$ back in and isolate μ .

$$P\left(-z_{\alpha/2} < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < z_{\alpha/2}\right) = 1 - \alpha$$

Multiply all three parts by $\sigma/\sqrt{n}$:

$$P\left(-z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \bar{X} - \mu < z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

Subtract $\bar{X}$ from all three parts:

$$P\left(-\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < -\mu < -\bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

Multiply by -1 , which flips both inequality signs:

$$P\left(\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

The two ends are exactly the confidence interval $\bar{X} \pm z_{\alpha/2} \sigma / \sqrt{n}$.

Step 5: what the probability belongs to. The last line looks like “ μ lies in this range with probability 95%”, but μ is a fixed number: the true mean age of the subscribers does not vary. What varies is $\bar{X}$: the next online event brings different people and a different mean. The probability belongs to the random interval, not to μ .

So once we compute one interval from one sample (25.06 to 30.94), we do not say “ μ is in it with probability 95%”. We say we are **95% confident**: 95% of intervals built this way contain μ . The [interpreting confidence intervals Note](#) shows this with a simulation.

9. Finding the critical value $z_{\alpha/2}$

Key point: Look up the area $1 - \alpha/2$ in the z-table: 0.975 gives $z = 1.96$ for 95%, and 0.875 gives $z = 1.15$ for 75%.

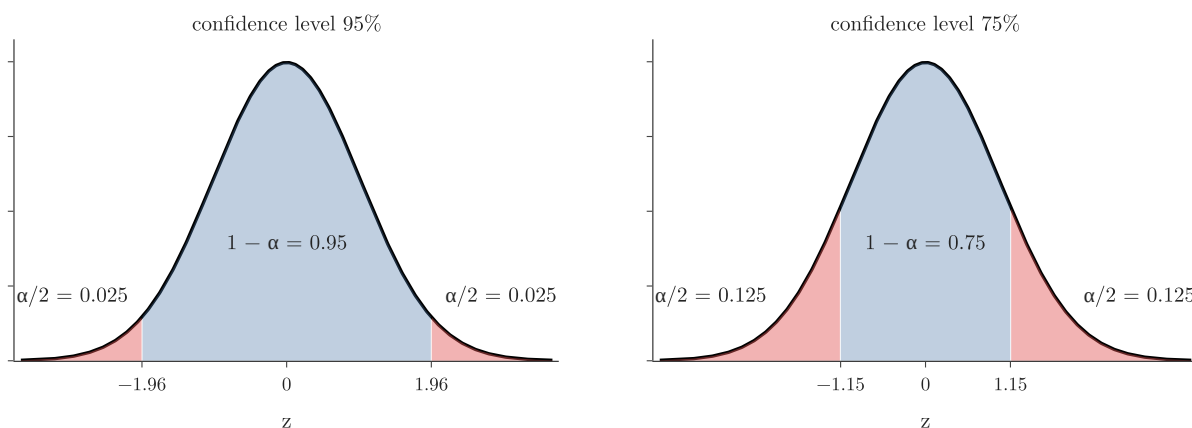


Figure 3: The middle $1 - \alpha$ of the standard normal curve and the critical values: ± 1.96 for 95%, ± 1.15 for 75%

A z-table gives the area to the **left** of z (see the [standard normal Note](#)). The area to the left of $+z_{\alpha/2}$ is the middle $1 - \alpha$ plus the left tail $\alpha/2$.

1. **In words:** add the left tail to the middle area, then find the z with that area to its left.

2. Formula:

$$\Phi(z_{\alpha/2}) = (1 - \alpha) + \frac{\alpha}{2} = 1 - \frac{\alpha}{2}$$

3. **Example:** for 95%, $\alpha = 0.05$:

$$\Phi(z_{0.025}) = 0.95 + 0.025 = 0.975$$

The z-table has 0.9750 in row 1.9, column 0.06, so $z_{0.025} = 1.96$. By symmetry the lower cut-off is -1.96 .

For a 75% confidence level, $\alpha = 0.25$ and each tail holds 0.125. The area to the left is $0.75 + 0.125 = 0.875$, which the z-table places at $z = 1.15$ (Figure 3, right). The 75% interval for the subscribers is $28 \pm 1.15 \times 1.5 = 28 \pm 1.73$: 26.27 to 29.73 years, narrower than the 95% interval.

Confidence level	$\alpha/2$	Area to the left	$z_{\alpha/2}$
50%	0.25	0.75	0.674
75%	0.125	0.875	1.150
80%	0.10	0.90	1.282
90%	0.05	0.95	1.645
95%	0.025	0.975	1.960
99%	0.005	0.995	2.576

That completes the z-procedure: with $\bar{x}$, σ , n and the critical value for the chosen confidence level, the interval follows from one formula. 95% is the usual level; the next Note shows why.

Python: Critical values with scipy.

```
from scipy import stats
```

```
level = 0.95
```

```
stats.norm.ppf(1 - (1 - level) / 2)    # 1.96
```

```
stats.norm.ppf(0.875)                  # 1.15, for 75%
```

ppf is the inverse of the CDF: it takes an area and returns the z with that area to its left.

10. Summary

Idea	Formula	Example
Confidence interval	point estimate $\pm$ margin of error	25 ± 4 : 21 to 29
Confidence level	$1 - \alpha$	0.95, so $\alpha = 0.05$
Standard error	$\sigma / \sqrt{n}$	$15 / \sqrt{100} = 1.5$
Critical value	$\Phi(z_{\alpha/2}) = 1 - \alpha/2$	$\Phi(1.96) = 0.975$
Margin of error	$E = z_{\alpha/2} \sigma / \sqrt{n}$	$1.96 \times 1.5 = 2.94$
Z-interval	$\bar{x} \pm z_{\alpha/2} \sigma / \sqrt{n}$	25.06 to 30.94 years

- A point estimate is almost never exactly right; an interval states the uncertainty.
- A confidence interval is about a population parameter, built from sample statistics.
- The z-procedure needs a random sample, a known σ , and a normal population or $n > 30$.
- The probability of 95% belongs to the method (the random interval), not to the fixed μ .

11. Key terms

Term	Meaning
Confidence interval	A range of plausible values for a population parameter, computed from a sample: point estimate $\pm$ margin of error
Confidence level	The share of intervals built by the method that contain the parameter, such as 95%; written $1 - \alpha$
Margin of error	The distance from the point estimate to either end of a confidence interval
Lower and upper limit	The two ends of a confidence interval
Z-procedure	The confidence interval $\bar{x} \pm z_{\alpha/2} \sigma / \sqrt{n}$, used when σ is known
Critical value	The z (or t) value that leaves $\alpha/2$ in each tail; 1.96 for 95% on the standard normal curve
α	One minus the confidence level: the share of intervals that miss the parameter